



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 1: AI & DS for Healthcare

Venue: CHT - I Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.S.Saranya

Sl. No.	Paper ID	Paper Title	Authors
1	48	IoT Based Gesture-Controlled Mouse System for Individuals with Hand Impairments	Aldarin Jino , Deva Priya Isravel, Julia Punitha Malar Dhas
2	101	Prediction of Epileptic Seizures Using Liquid Time-Constant Networks	Darshan Shestha , Saranya G, Padmanabha Sarma, T. Swethaa, Manjushri M
3	215	Detecting Schizophrenia through EEG Signals using Deep Learning	Ashish Make , Swarnalata Bollavarapu, Radhika Baid, Devansh Gupta
4	86	MINDCHRONO: A Comparative Study of Deep Learning Models for EEG-Based Depression Detection	Balakrishna N , M.B. Mukesh Krishnan, D. Ganesh
5	124	Adaptive Deep Knowledge Framework for Classifying Sleep Stage using Deep Feature Learning	Amoga Lekshmi S , Deepthi R, Amutha R
6	366	A Framework for Early Detection and Analysis of Prenatal Brain Development	Sathiyavathi S , G. Udhayakumar
7	243	A Medical Chatbot for Diabetic Patients Using Artificial Intelligence	Mangai S ,A.Supriya,V.Kavitha
65	117	Enhanced MRI-Based Brain Tumor Detection Using Deep Learning Architectures and Optimized Machine Learning Techniques	Narenthirakumar Appavu



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 2: Biomedical Imaging and Image Processing

Venue: BMI - Ground Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr. K. Nirmala

Sl. No.	Paper ID	Paper Title	Authors
8	123	Diabetic Retinopathy From Retinal Imaging	Soorya , Narayanan V, Mohammed Anfaal D, Rajadurai S, Akash V, Divya Bharathi S, Sandhya C
9	79	Particle Swarm Optimization of CNN for Early Parkinson's Disease Detection	Sharvesh T , Srinivasan, Shrivan-shi Ganesh Pai, Basavaraj Hiremath, Ramsha Fathima, Parthib Basu
10	52	Analysis of Blood flow in Carotid Artery at Various Blood Viscosities During Rest and Physical Exercise	Sankar Kandasamy , Kalpana, Ramakrishnan, Malathi M
11	370	Deep Learning-Based Automated Breast Cancer Ultrasound Image Classification: A Study	R Sivakumar
12	157	Transfer Learning with Deep Convolutional Neural Networks for Esophageal Cancer	Susendhiran M , Akshaya A, Jeevitha S, Gowri S
13	158	Hybrid Ensemble Methods for Acute Lymphoblastic Leukemia Diagnosis: Integrating CNN Features with Artificial Rabbits Optimization Algorithm	Divyadharshini K , Suvetha R, Alagu S, Harithra R
14	374	Image Enhancement using linear filters for Rectal Cancer Diagnosis using MR Images	Sonal patreena D , Almeida, Surekha Kamath, Rajgopal KV, Manjunath KN
66	119	Deep Learning for Predicting Invasive Ductal Carcinoma in Histopathological Tissue Images	Narenthirakumar Appavu



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 3: Biomedical Sensors, Instrumentation and Wearable Systems

Venue: DTE Lab II Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.Suganthi L
Student POC: Ms.B.Madhumitha

Sl. No.	Paper ID	Paper Title	Authors
15	361	Thermogravimetric Analysis and Tensile Testing of PEEK Structures Fabricated via Extrusion and 3D Printing	Marieswaran M , Ravi Tiwari, Somnath Bhattacharya, Sanjeev Kumar
16	244	Software for Clinical Speech Therapy Services	Mangai S, S.Govindaraj,K.Anjana
17	352	A Smart Shoe for Detection and Alleviation of Foot Ulcers	Sri Balaji P , Vishnu N, Sai Mathangi S, Dr. S. Poonguzhali
18	126	A Study of Spectral Preprocessing Techniques for Bilirubin Quantification Using DRS	Chitra R , Shanthi Prince
19	149	Self-Stabilized Gyro Toothbrush For Parkinson's Disease: Enhancing Oral Hygiene Through Real-Time Motion Compensation	B. Nivethitha , A. Kirubhashini, A. Josephin Arockia Dhivya, Jaya Rubi
20	142	Terahertz-Based Biosensor for Characterization of Cancerous Cells	Hemalatha E , Lalit Dhakshan M, Karthick R, Meraline Selvaraj, Dr. Sreeja B S
21	164	OptiVue - A Smart Approach to Eye Health Screening	Swetha A , Kiruthikha M, Ramyaa G R, Dr. S. Prabhakar, Sivagowrika S
78	121	Improved Retinopathy Detection Using ResNet-50 Optimized AI Deep Learning Models	Narenthira Kumar Appavu



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 4: Biomedical Signal Processing

Venue: MSDM Lab I Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.B.Geethanjali

Sl. No.	Paper ID	Paper Title	Authors
22	104	VLSI Architecture for Modulating Nerve Activity in Parkinson's Disease Using STFT Transformation Methods	S. Karthick , Neelamegam Devarasu
23	19	AI Powered Heart Rate Monitoring System Leveraging Skintone	Kanimozhi V , V. P. Kalaiyarasi, Danish Mithula S, Kishore I M
24	319	Waves Don't Lie: Leveraging Test-Time Training and Kolmogorov Arnold Networks for EEG-Based Biometrics	Eshwanth Karti T R , Nithish Ariyha K, Vikash J, Yeshwanth Balaji AP, Amrutha Veluppal
25	118	Evaluation Methods for Studying Muscle Fatigue through Physiological Signal Monitoring	Annapoorani CL , Sofia Bobby J, Subha Ramya, Miruthula A, Mythily V
26	248	Preterm Birth Prediction Using GraphSAGE and Electrohystero-gram Data	Nandhini Kuppala , Monish GV, Nivedha V, Balu Pinnenti, Amrutha V
27	278	ESP32: Smart Support Empowering Alzheimer's Patients (On-line)	Aloysious Alex , Alen Christy C R, Adityan Ramesh, Hridhya K H, Anju Babu, Siji Joseph
28	176	MaskCycleGAN-based Whisper to Normal Speech Conversion	Johanan Joysingh S , Rohith Gupta K, Ramnath K, Vijayalakshmi P



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 5: Materials for Bioscience

Venue: BME Seminar Hall III Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.R.Subashini

Sl. No.	Paper ID	Paper Title	Presenter Name
29	156	Predictive Analytics in Dentistry: AI Applications in Frenectomy and Soft Tissue Management (Online)	Ashna Yuvaraj , Pradeep Kumar Yadalam, Karthickraj S, M. Soundharya Manogaran, Swarnambiga
30	228	Segmentation Of Neutrophil Nucleus in White Blood Cell Images Using Kapur's Entropy-Based Improved Grey Wolf Optimization	Saranya R , Bakiya A
31	154	Machine Learning Prediction of Marginal Gingival Adaptation of Direct Composite Veneering with or without Mock-Up in Upper Anterior (154)	Ashna Yuvaraj , Swarnambiga Ayyachamy, Pradeep Kumar Yadalam, Soundharya Manogaran, Karthickraj S.
32	78	Indian Herbal Formulation for Repigmentation of Vitiligo	K. Mythili , S. Prabakar
33	42	Incorporation of Banana Leaf Extract into Collagen Sheets for Fire Burn Healing	Brindha M , Nareshkumar N, Prabakar S, Nathish B, Dharanya S, B. Poovarasam
34	321	Design of Components for a 3D Printed Assistive Robotic Arm Optimized for Precision Surgical Procedures	Jaya Rubi , T. R. Thamizhvani, R. J. Hemalatha, B. Varshini, A. Josephine Arockia Dhivya, S. G. Viena
74	225	A framework of a smart therapeutic insole system by measuring plantar pressure for early diagnosis of diabetic foot ulcers using machine learning and thermal imaging techniques	Diviya S , Sangamitra N, Venuhasini S M, B. Suresh Chander Kapali, A. Saravan
70	314	Myoelectric Controlled Lower Limb Prosthesis with Active Ankle-Foot Joint	M. Neela Harish , Harini Sri Kamatchi K, S. Sathish, S. Prasanthi



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 2 - Track 1: AI & DS for Health Care

Venue: CHT-I Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.S.Saranya
Student POC: Ms.Shrinithi SK

Sl. No.	Paper ID	Paper Title	Presenter Name
35	87	A Novel & Hybrid Algorithm for Real-Time Depression Detection via EEG Analysis	N. Balakrishna , M. B. Mukesh Krishnan, D. Ganesh
36	181	Wheat Disease Detection: Bridging the Gap with Deep Learning Approaches	Vijaykumar Vasaiya , Amit Vajpayee, Ankita Gandhi
37	165	Improving Hepatitis-C Detection Accuracy Using Improved C5.0 Algorithm with Chi-Square Feature Selection	Shamitha G , S. Alagu, Sushmitha A, Shohail Shahabudeen
38	178	Southern Regional Healthcare Chatbot Using AI and NLP	S. Raja Mohamad , Muthukamalesh V, Sreeju S. A, Akaash Henry V
39	138	AI-Based Decision Support System for Body Constitution (Yakkai Ilakkanam) Classification Using Minimal Optimized Textual Features	Keerthivasan G , Sutha S, Santhosh K M, Pappa N, Praveen Kumar L
40	258	A Study On Cardiovascular Disease Detection In Diabetic Patients	Keerthana S. , G. Anitha
41	344	Chronic Kidney Disease Prediction Using Deep Learning	Hepsiba D , Vignesh R



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 2 - Track 2: Biomedical Imaging and Image Processing

Venue: BMI - Ground Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.R.Nithya

Sl. No.	Paper ID	Paper Title	Presenter Name
42	281	Automated Deep Learning System for Early Detection of Stroke Using CNN Architecture	Kshnika O , S. Gunasekar, G. Babu, M. Neela Harish
43	332	Computational Linear Modeling for Human Lung Mechanics	L. Priya , Vishnupriya B, Dhanusree N, Yogeshwaran S, Priyadharshni S
44	72	Prosthetic Hand Grasp Recognition Using Residual-Unet Combined with Cross Over Attention	Premalatha G , Srikanth Vasamsetti, Srivatsal Narayan Sekar, Anup Chander
45	373	Early Detection of Liver Disease and Precise Health Optimization Using Advanced Computational Models	Rubin Bose S , Angelin Jeba J, S. Viswanathan, V. Mahalakshmi, V. Hari, Regin R
46	153	Detection of Finish Lines in Indirect Ceramic and Composite Veneers Using Clinical Images with the Help of Developed AI-Based System (Online)	Ashna Yuvaraj , Soundharya Manogaran, Pradeep Kumar Yadalam, Swarnambiga Ayyachamy, Karthick Raj
47	50	Automated Pneumonia Identification Using Convolutional Neural Network	Gnanapriya M , V. Loganathan, S. Prabakar, Revathi N, Rakshagamoorthy V, Karthiswaran M
48	76	Multimodal Analysis of Hereditary Breast and Ovarian Cancer Syndrome Using Deep Learning Techniques	Dhivya Pradeep , Subham Lakshmi B, Shanthini Devi P, Kavitha G,



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 2 - Track 3: Biomedical Sensors, Instrumentation and Wearable Systems

Venue: DTE Lab II - Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.Subitcha J

Student POC: Ms.Ezhil Meenaa A S

Sl. No.	Paper ID	Paper Title	Author Name
49	325	Ergonomically Optimized Wearable Seating Posture Monitoring and Alert System for Mitigating Work-Related Musculoskeletal Disorders	Christopher Dennis J , Raksana B, Rakshita T S, Dr. Poonghuzhali S
50	166	Flexural Transducer Based Monolithic Diathermy Probe for Thermo Therapy	Nithyakalyani K , Nanthini B, N. Bharathiraja, Vishnupriya A, Madhumitha K, Harinni K K
51	167	CARE-DAQ: Low Cost Open Source Wireless Biomedical Data Acquisition System	Ramakrishnan Maharajan
52	312	Real-Time Health Monitoring and Tracking System	Shelma Vincent V. S , Hepsiba D, Amala Hepzibha D, L. D. Vijay Anand
53	120	Self Powered Energy Harvesting Wearable Monitor	Sofia Bobby J , Mythily V, Annapoorani C. L, Reena Shree, Subha
54	323	Smart ICU Revolutionizing Critical Care with Multi-Modal Sensing and MIoT-Driven Solutions	Iyyanarappan A , Shakira Banu S, Madhumitha B.
55	111	TheraLink Smart Health Multiparametric Monitoring System with Wireless Data Transmission	Rohan Steven B , Anne Joan Benita V, Soorya Narayanan V, Sandhya C, S. Vinurajkumar, Divya Bharathi S



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 2 - Track 4: Biomedical Signal Processing

Venue: MSDM Lab I Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.M.Dhanalakshmi
Student POC: Ms.Soundharya

Sl. No.	Paper ID	Paper Title	Author Name
56	233	Machine Learning Based Categorization of Heart Sounds for Cardiac Arrhythmia Detection	Sreedevi Nair , Indu T
57	330	Biophysical and Anthropometric Determinants of Muscle Co-activation Patterns during Human Gait	Arunthathi Sivalingam , Saranya S
58	303	HealthNet: Secure and Scalable Hospital Network for Enhanced Patient Care	M. Vivek Kumar , K. Sureka, S. Shakthi Poovannan, M. Thirisankari, V. Soundharya
59	324	Using Web Mining to Track and Predict the Spread of Diseases (Online)	Sadik Khan
60	342	Single-Modality Emotion Detection: EEG-Based Feature Engineering and Interpretability	Vishal A S , Deepan S, Amrutha V
61	322	Realtime Cardiac Health Monitoring Using AI-Powered Embedded Systems	Tarun V. M , Madhumithaa S, Masoodhu Banu N. M, Jehosheba Margaret M
62	43	Leveraging Spectral Signatures and Attention Enhanced BiLSTM Network for Comprehensive Respiratory Sound Analysis	Rohit VP , M V V S Swaroop, Mithin Dev A, Suriya KP, Amrutha Veluppall



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 3 - Track 1: AI & DS for Health Care

Venue: CHT-I Floor, BME, 11:15 p.m. – 12:30 p.m. IST

Internal Session Chair: Dr.J.Vijay
Student POC: Ms.Gayathri

Sl. No.	Paper ID	Paper Title	Presenter Name
63	175	Detecting Childhood Autistic Spectrum Disorder and Enhancing Their Reading Skills	Bino J , Pratheeksha S, Shameera S I.
64	180	Machine Learning in Mental Health: Bridging Gaps in Diagnosis and Intervention (online)	Nirav Kansara , Brijesh Vala, Mohammad Shahnawaz Shaikh
67	193	Voice Assistance for Visually Impaired Persons Using AI and IoT (online)	Ramesh L , Pruthiga M S, Muthulakshmi M, Puvitha Sri N, Navaneetha S
68	174	An Intelligent System for Heart Sound Signal Analysis: A Survey	Manu Saraswat , Chhote Lal Prasad Gupta
69	155	Evolution Scale Modelling for Prediction of Regulator G Protein Signalling (RGS) Sequences in Oral Cancer	Soundharya Manoharan , Swarnambiga Ayyachamy, Ramya Ramadoss, Pradeep Kumar Yadalam



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Day 3 - Track 2: Biomedical Imaging and Image Processing

Venue: BMI Ground Floor, BME, 11:15 p.m. – 12:30 p.m. IST

Internal Session Chair: Dr.S.Suresh Anand

Student POC: Ms.Nithiyaa Dharshini

Sl. No.	Paper ID	Paper Title	Presenter Name
71	349	An Advanced Smart Glove for Stroke Rehabilitation Using IoT	Dharun R
72	274	Simulation and Experimental Study of Quartz-Enhanced Photoacoustic Spectroscopy Technique for Human Exhale Breath Gas Sensing	Saran Krishnamoorthy , Saran Kumar, Kishore S, Sooraj Shiby, Nilesh J Vasa
73	313	Injectable Hydrogel Loaded with Copper Oxide–Curcumin Nanosonosensitizer for Enhanced Antibacterial Sonodynamic Effect	Abiramy Soundararajan , Preethi Vasundara N, Bhavya V, Akshaya A. P.
75	182	A Deep Learning Approach to Multi-Class Skin Disease Identification	Hariharan R , Dhanushraju S, P. Dharshini Muthu Bala, Saaral S, Devi Vijayan



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 3 - Track 3: Biomedical Sensors, AI, Instrumentation and Wearable Systems

Venue: MSDM Lab, I -Floor, BME, 11:15 p.m. – 12:30 p.m. IST

Internal Session Chair: Dr.S.Arun Karthick
Student POC: Ms.Aruna Devi

Sl. No.	Paper ID	Paper Title	Presenter Name
76	195	Hybrid Deep Learning for Detecting Fake Medical Images and Heart Sounds	Soundarya , R. Vijayalakshmi, P.Priya Dharshini
77	230	Intelligent Intravenous Drip Stand for Safe Infusions Using IoT Integration	Sushma Philipraj , Remya Raj
79	249	Hybrid Deep Learning Models for Early Detection of Chronic Kidney Disease: A1-D CNN and ResNet 152-V2 Approach	Jeena Jose , E. J. Thomson Fredrik
80	256	Harnessing Deep Learning for Thyroid Disease Detection: A Systematic Review	Fasira T , S. Jerine
81	341	EEG-Driven Massage Headband for Anxiety Alleviation (Online)	K. Sravani , Dr. Swathi Nadipineni, B. Anjali, Ch Vedavathi, B. Ramya
82	336	Histopathological Image Analysis Using Improved Gorilla Troops Optimizer for Lung Cancer Detection	Bharanidharan N , Nepolian V